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Ministry of Health
The Royal Hospital
Department of Child Health

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Title: Guidelines of the Management of Central Venous Catheters related blockage for Children

The use of central venous catheters(CVC) has become an essential part of care for Children with critical illnessor those with chronic diseases.

This guideline describes how toprevent CVC blockage and how to unblock blocked lumens.

This guideline is to be used in thepediatric wards and critical care areas.

1. Preventing central venous catheters blockage:

1.1:Thefollowing steps are advised to avoid CVC blockage:

1. Use of patency devices to avoid line thrombosis. Ifused, patency devices should be Changed every 7 days.
2. Use appropriate amount of 0.9% saline following viscous and non- viscous solutions, bloodcollection and withdrawal of blood. (Refer to Tables 1 and 2)
3. Follow the Push and Pause/ turbulent orpulsatile flushing techniques and thepositive pressure technique for flushing the CVCs. Pulsatile technique helps toremove the residuals of medications or blood that adheres to the internalsurface of the catheter. While, the positive pressure technique ensurescreation of a pressure inside the catheter that prevents back flow of the bloodinto the catheter lumen.
4. Use 10 ml syringes to flush the centrallines after use.
5. Alternate the use of different lumens forblood sampling. Use the wider lumens for blood sampling.

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	Types of Central Venous Catheters				
Saline used	**Hickman (tunneled cuffed)**	**PICC (Open Ended)**	**PICC (Valved)**	**Multiple lumen non tunneled short term catheters**	**Implanted port**
Before Medications	**1-2 ml**	**1-2 ml**	**1-2 ml**	**1-2 ml**	**2-3ml**
In between Medications	**1-2ml**	**1-2ml**	**1-2ml**	**1-2ml**	**2-3ml**
After Medications	**1-2ml**	**1-2ml**	**1-2ml**	**1-2ml**	**2-3ml**
After Blood Sampling	**5 ml**	**5 ml**	**5 ml**	**5 ml**	**5 ml**
After Blood Administration	**5 ml**	**5 ml**	**5 ml**	**5 ml**	**5 ml**

Table 1. Recommended flushing volumes of normal saline 0.9% before, in betweenand after medications' administration, blood sampling and administration.

Pleaseconsider the child overall fluid state.

It is a must that the pulsatile and positive pressure techniques areused for flushing of CVC.

*Please note that higher volumes of saline flush of5-10 ml can be used after viscous medications, after blood sampling and bloodadministration in children where volume status is not restricted.

*To prevent peripheral lines blockage, Use continuousinfusion 0.9% saline at rate of 1 ml/hour for a neonate and 2 ml/hour forinfants and older children when small lines are not in use.

Line use < 6 hours Short term lines(percutaneous lines) Use continuous infusions or flush line every 6 hours	Use sterile normal saline as indicated above
Line access 7-24 hours	Use short term heparin lock (10 iu/ml)
If line accessed >24 hours All tunneled (non- cuffed) and PICC lines need daily flush. Hickman/ Broviac need flush and re- lock every 7 days. Implanted ports can be flushed every 4 weeks	Use long term heparin lock (100iu/ml) ** It is a must to aspirate heparin lock first and then flush with 5-10 ml of normal saline using pulsatile flush

Table 2: recommended type of flushes and frequencyof flushes for CVC

2. Treating blocked lumens:

In cases of lumen obstruction the following steps are advised:

- Assess the site for redness, tenderness and swelling.
- Flush the lumen with 0.9% normal saline to check the patency of thecatheter.
- Aspirate each lumen to determine if easy to flush and aspirate.

2.1: Types of lumen obstruction

The line Obstruction can either be secondary tomechanical obstruction or thrombotic occlusion.

Causesfor mechanical obstruction can be the followings:

- * Kink of the catheter.
- * Tight sutures
- * Clamped lumens
- * Catheter malposition or internal kinking of the catheter

Thrombotic obstruction of the linecan be 2nd to intra-luminal obstruction with fibrin.

Typesof line occlusion:

- * **Partial:** decrease ability to infuse fluids or resistance while infusingfluids.
- * **Withdrawal**: ability to infuse through the line without ability to have freebackflow.
- * **Complete:** inability to withdraw or infuse fluids or blood.

All types of line occlusion need treatment.

Clinical manifestations:

The child with catheter blockage can manifest with the followingsigns:.

- Discolorationof limb, paresthesia or loss of function.

2.2: Managementof thrombotic obstruction of a lumen.

The followingsteps to be followed:

1. Flushand aspirate the lumen to make sure about patency.
2. Checkbackflow with small syringe (1ml or 3ml) as it has less negative pressure withdrawing.
3. Donot try to flush the lumen with small syringe because of high positive pressure
4. If the lumen is blocked administer thrombolytic agent (see table3)

Notethat The catheter can also become blocked by chemical occlusion.

2.3: Instillationof the thrombolytic agent and the required dwell times

Please consult the most senior physician (consultant) before requesting for thrombolytic agent.

The thrombolytic agent should be instilled and left to dwell.

The thrombolytic agent should be aspirated after dwell time.

It is recommended to instill up to 2doses of alteplase (t-PA) to fill up the catheter lumen dead space (the primingvolume of the catheter).

Referto next table on different priming volumes for commonly used catheters.

CVC Type	Size	Priming Volume
Hickman (Broviac) ***Product Brand: Bard***	2.7 Fr	.3ml
Hickman (Broviac) ***Product Brand: Bard***	4.2 Fr	.4ml
PICC (open ended) ***Product Brand:MedComp***	4 Fr (20cm)	.4 ml
PICC (open ended) ***Product Brand:MedComp***	3 Fr	.3 ml

Product Brand: Cook	5.5, 6.5 &7.5	4 ml (Based on the product guide: **5ml**)

Note: The overall priming volume of the CVC must not exceed 0.5 ml. Pleasecheck the intervention radiologist notes prior to installment of the primingvolumes.

For shortterm catheters, the priming volumes for different lumens are found on theproduct package.

1. Keep the dwelling time forthrombolytic agent 30- 120minutes for first dose.

2. In case of catheter is not fullypatent, repeat 2nd dose for 30 minutes can be administered

3. Instillalteplaseby direct-syringeinstillation method if partial or withdrawal occlusion (appendix 1)

2.4: assessmentof the catheter patency:

Assesspatency of the catheter by ability to draw 3-5 ml of blood following t-PA dose.

Otherchoices for thrombolytic agent:

Reteplasecan also be used for unblocking of blocked lumens.Please refer to the appendixon the use of Reteplase as Thrombolytic agent.

Appendix1(methods for instillation for thrombolytic treatment)

Thereare two methods or techniques to create negative pressure for drug instillation

1. **The single-syringe technique**

   ➢ Use a single 10 mL syringe

   ➢ Attach directly to the occluded lumen hub.

   ➢ When the plunger is pulled back a vacuum is created.

   ➢ The plunger is then slowly released; this allows the thrombolyticto

   be “pulled”back into the lumen toward the thrombus/clot

2. **Three-way stopcock attached to the blocked lumen,**

   ➢ Attached to the three way an empty syringe in one port and sterile10 mL syringe to other port.

   ➢ The plunger of the empty syringe is pulled back to create a vacuum

   ➢ Keep the stopcock off to the empty syringe and open to thethrombolytic syringe to allow the drug to be “pulled” into the catheter to theocclusion

   **note: withboth techniques, the syringe plunger will need to be pulled back several timesto permit full instillation of the drug.

3.1: Avoiding lines related blockage

- * CVC in neonatalshould be always kept with continuous infusion minimum of 1ml/hr.

- * If a centralline (CVC) is not being used for continuous administration of fluids, it shouldbe flushed six hourly with a solution containing heparin per ml (1IU/ml).

- * PICC linesshould always have a minimum of 1ml/hr I.V Fluid running through them toprevent blockage.

  * RemovePICC line catheter as soon as it is no longer medically necessary

- * Use continuousinfusion of 1ml/hr of I.V Fluid for maintenance of patency of central venouscatheters and umbilical venous catheters.

Intermittentheparin flushes:

Long lines, UACs and UVCs are to be primed with and flushed with0.9% saline during insertion. After insertion, connect to continuous infusiondepending on fluid needed for the baby.

3.2: Flushingof central lines:

- * When using aCVC, bolus injections must be slow and preferably using a 10 ml syringe

- * The line shouldbe flushed with intravenous saline 0.9% or 0.45% using the larger syringe sizeinitially to determine patency. (The central venous catheters are at risk of rupturingif the infusion pressure exceeds 25 psi, which may happen using a 1 or 2 mlsyringe).

- CVC: It isnormal practice in the NICU to use a syringe driver or an IV infusion pump toinfuse any fluids into any venous access device.

- * If flushing byhand it is recommended that CVCs are flushed using a push-pause method tominimize the mixing of incompatible medications or fluids and to reduce fibrinbuildup or accumulation of medication precipitate in the catheter lumen. Thepush-pause method (stop- start) keeps the solution moving along the walls ofthe catheter. Do not flush rapidly as this can cause rapid changes incirculatory volume and pressure.

<u>When to flush a CVC:</u>

A CVC shouldalways be flushed before and after:

○ Administrationof medication

○ Administrationof blood and blood products

○ Intermittenttherapy

○ Blood samplingor withdrawal

3.3:unblocking of the line:

Central Venous Catheter(CVC)

- If the CVCbecomes blocked a member of the medical team, or critical care nurse can flushthe line using an aseptic technique if competent to do so.

heparinized saline (1 IU/ml) using a 10mlsyringe. This should be done with a gentle pressure.

- If able toflush line but no blood return, arrange X-Ray to confirm line placement andabsence of kinking
- If unable toflush obtain surgical consult and consider removal of the line

Peripherally Inserted Central Catheter (PICC) Line

- * For a PICCline, do not aspirate back, as blood will occlude the narrow lumen, but flushwith 1-2 ml of heparin in 0.9% or 0.45% saline solution (1 IU/ml) using a 10 mlsyringe. Do not flush excessheparin saline solution in,just enough to remove the blockage.

  * RemovePICC line catheter as soon as it is no longer medically necessary

- * Continuousinfusion: 1ml/hr of I.V fluid for maintenance of patency of central venouscatheters and umbilical venous catheters.

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